

Uncertainty-Aware Distributed RME in Decentralized Wireless Networks

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Outline

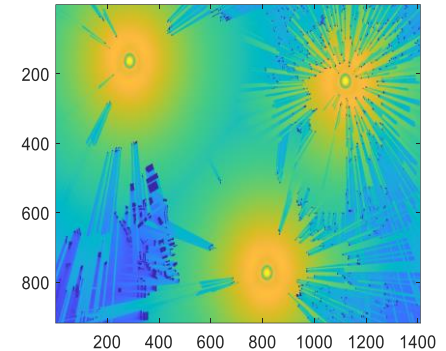
- ❑ Background
- ❑ Problem Formulations
- ❑ Introduction of Uncertainty-Aware D-RME
- ❑ UA-D-RME: The Proposed Method
- ❑ Experiments
- ❑ Conclusion

Background

❑ Radio map Estimation :

Estimating received signal strength(RSS) of the target area

- ❖ Model-based methods
- ❖ Data-driven methods
- ❖ Physics-inspired learning methods



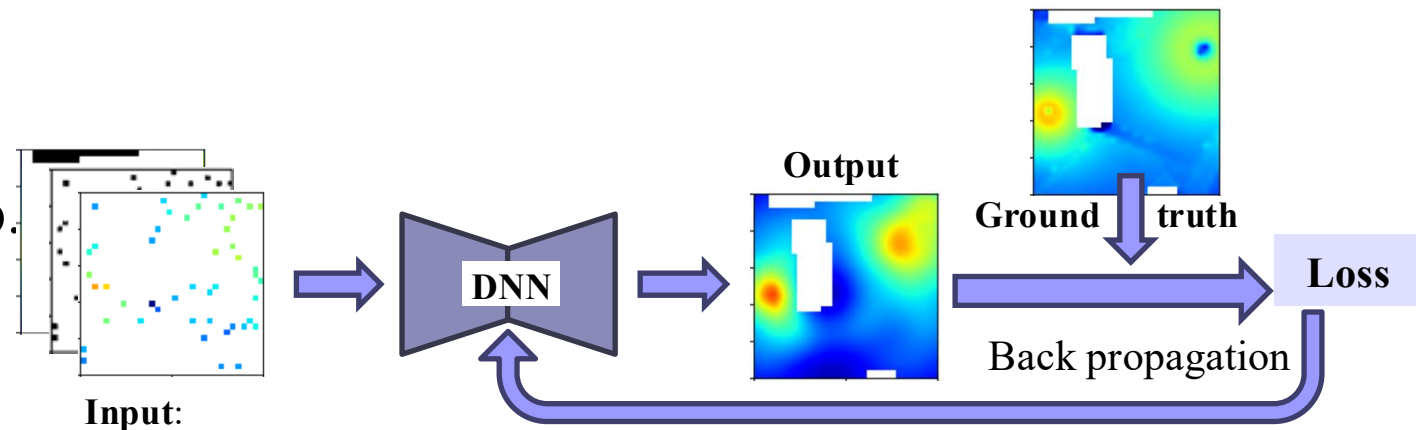
❑ Data-driven Methods (Centralized):

Generative DNN: Autoencoder, Unet

- From Tx configuration, building map.
- From centralized sparse sensor data.

Challenges:

- Data scarcity and overfitting
- Privacy concern



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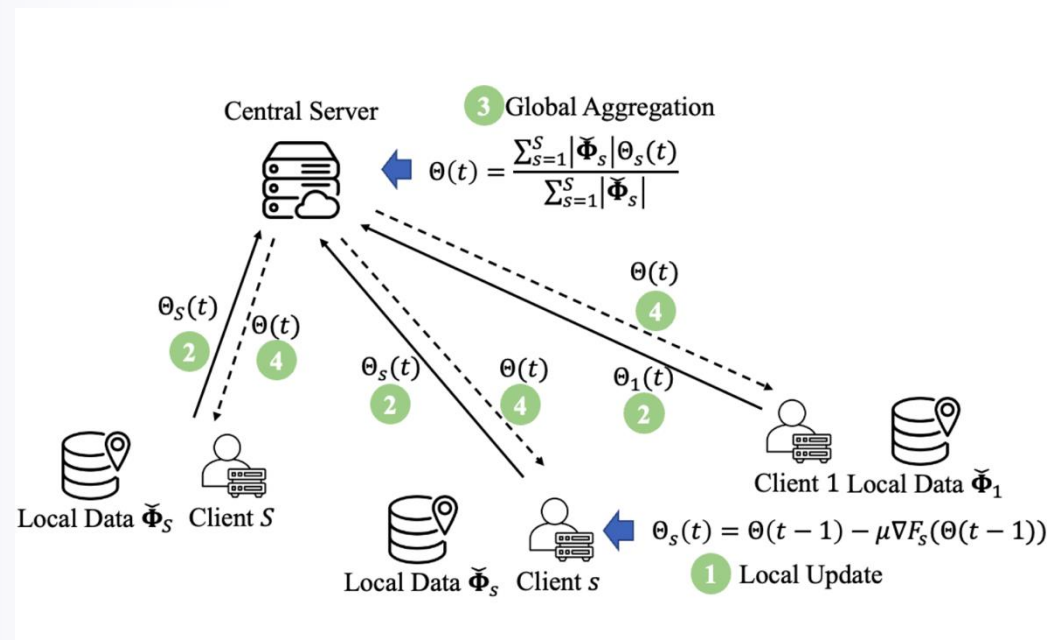
Problem Formulation of Distributed RME

□ Reconstruction loss:

$$F_s(\Theta) = \|\mathcal{O}_s \odot (g(\check{\Phi}_s; \Theta) - \Phi_s)\|_2^2 + \lambda \mathcal{R}(\Theta),$$

$$\Theta_s(t) = \Theta(t-1) - \mu \nabla F_s(\Theta(t-1)),$$

$$\Theta(t) = \sum_{s=1}^S \alpha_s \Theta_s(t) = \frac{\sum_{s=1}^S |\check{\Phi}_s| \Theta_s(t)}{\sum_{s=1}^S |\check{\Phi}_s|},$$



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Introduction of Uncertainty-Aware D-RME

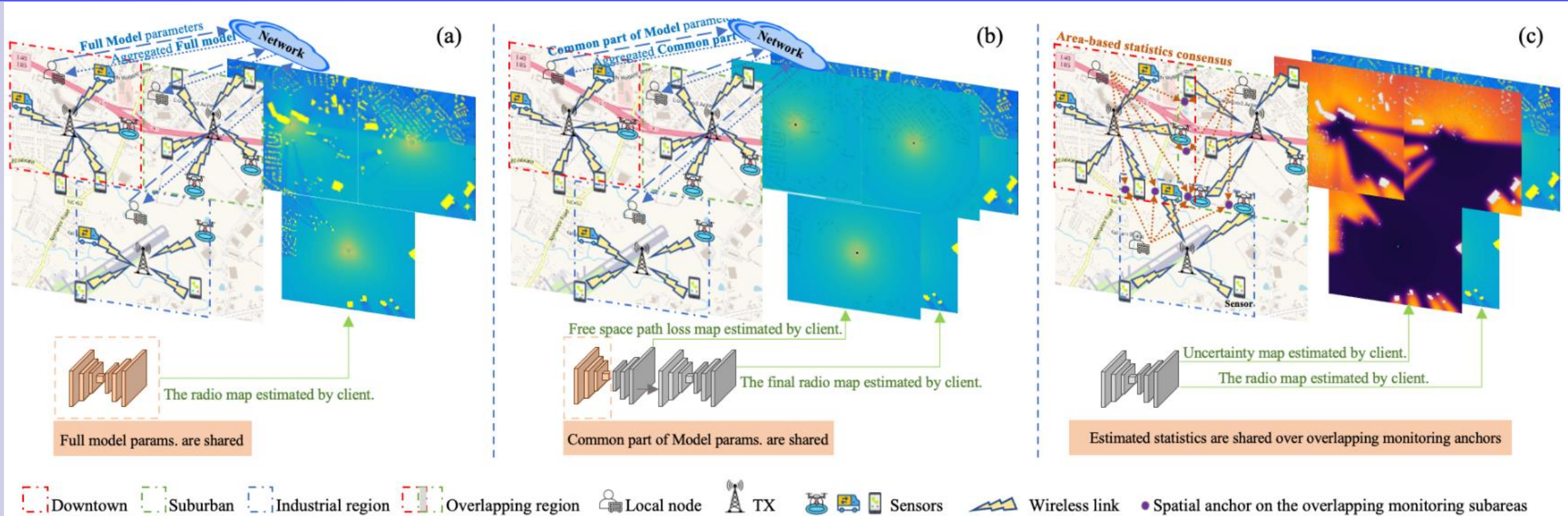


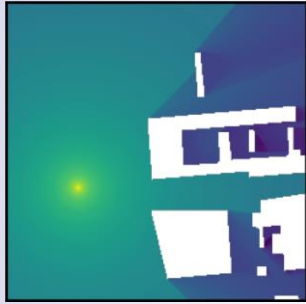
Fig. 1. Three distinct distributed radio map estimation frameworks: (a) A federated learning-based framework, where nodes update their local models using aggregated model parameters or gradients received from a central server, train them further on local data, and then transmit the results back to the central server; (b) A distributed framework via physics-inspired model decoupling, in which the model is decomposed into a global module and a client-specific module, with only the global module being shared at the central server; and (c) An uncertainty-aware distributed framework, where each node maintains a local probabilistic estimator and exchanges only low-dimensional predictive statistics regarding overlapping monitoring anchors.

How can distributed nodes collaboratively estimate radio maps, without exchanging raw measurements or massive model updates?

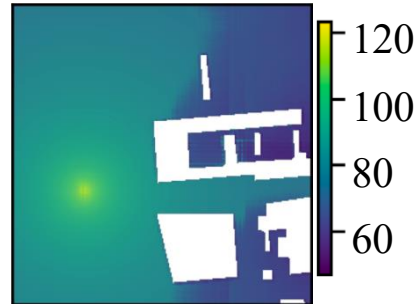
[1] Weishan Zhang, Yue Wang, Lingjia Liu, Zhi Tian. "FedRME: Federated Learning for Enhanced Distributed Radiomap Estimation." In VTC2024-Fall, 2024.

[2] Dong Yang, Yue Wang, Songyang Zhang, Yingshu Li, Zhipeng Cai. "Physics-Inspired Distributed Radio Map Estimation", In ICC 25, 2025.

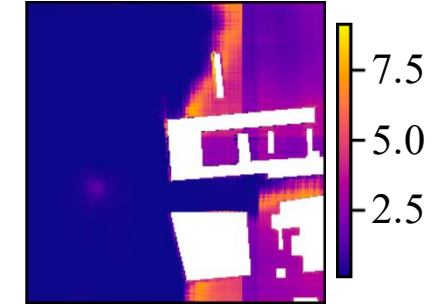
Uncertainty-Aware for Distribute RME



Ground Truth



Estimated RM



Uncertainty Map

Two insights

- **Uncertainty awareness capabilities** should be introduced to reduce divergence.
- **Consensus mechanisms** drive the realization of decentralization, rather than simple aggregation.

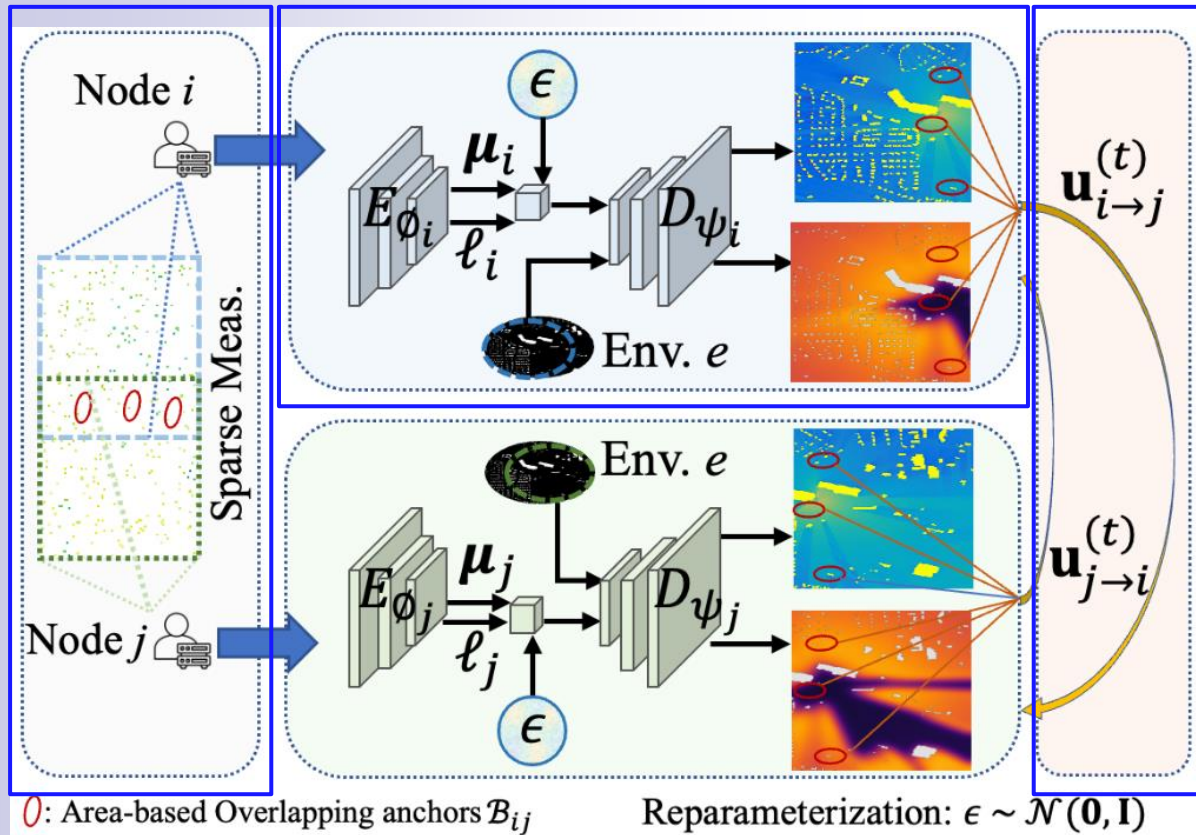
Uncertainty-Aware D-RME

- **Local Probabilistic Model** estimates the signal values and their corresponding uncertainties in the interest of region [3].
- **UA Statistical Consensus mechanisms** enables nodes to achieve RME consensus with their neighbors under communication constraints.

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Overview of Proposed Uncertainty-Aware D-RME



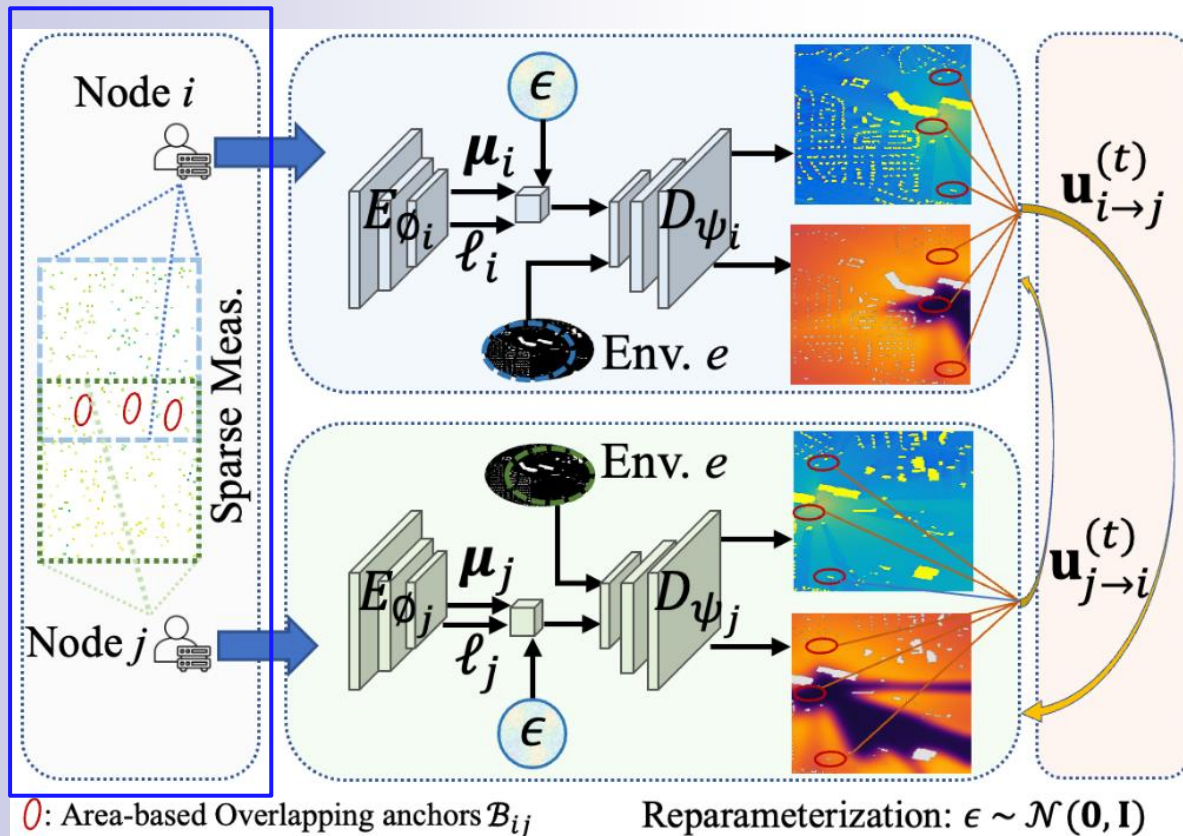
1 Distributed Local Sensing

2 Local Probabilistic Model

3 UA Statistical Consensus

UA Statistical Consensus achieves distributed RME under communication constraints.

UA-D-RME: Distributed Local Sensing



$\mathcal{A} = \{1, \dots, K\}$ distributed mobile edge nodes

Each node $i \rightarrow$ a local region $\mathcal{X}_i \subset \mathcal{X}$

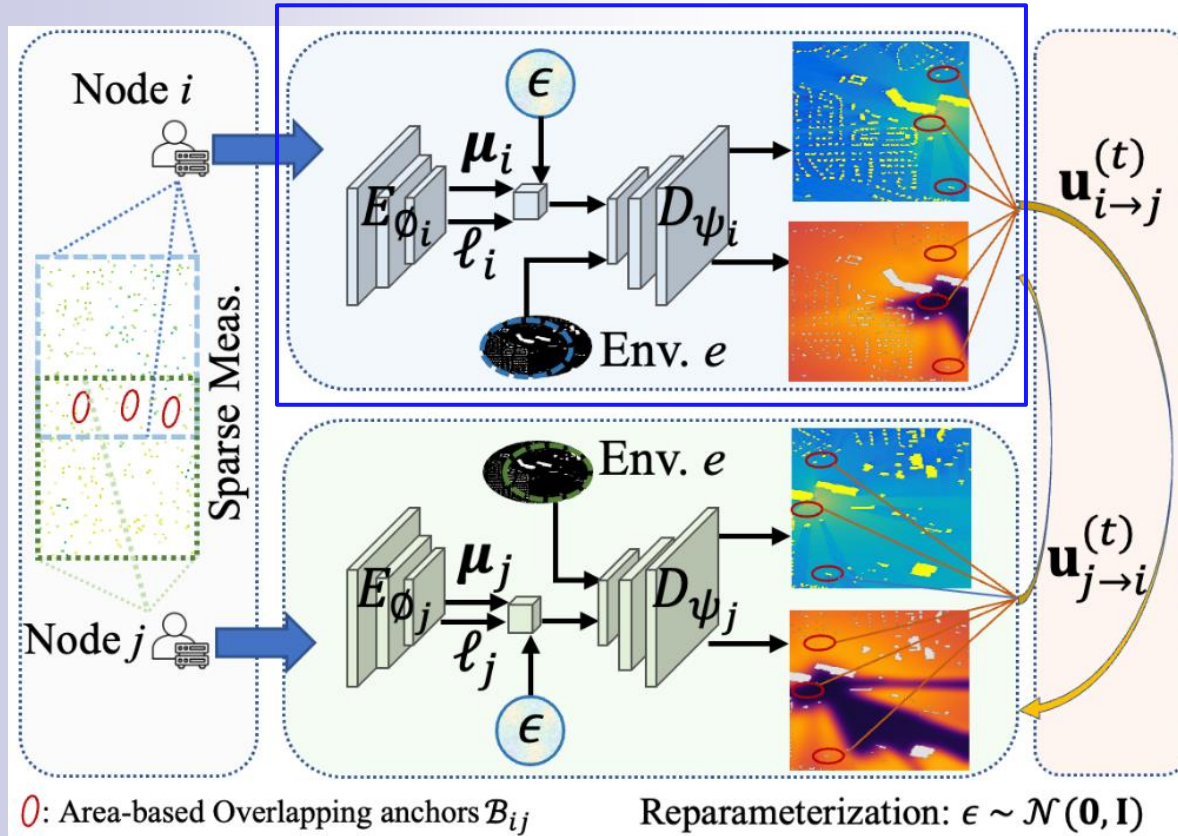
For any two adjacent nodes $\left\{ \begin{array}{l} \text{node } i \\ \text{node } j \end{array} \right\} \rightarrow$ **Overlapping area**
 $\mathcal{X}_{ij} = \mathcal{X}_i \cap \mathcal{X}_j$

Subareas $\mathcal{B}_{ij} = \{\mathbf{b}_{ij,q}\}_{q=1}^Q \subset \mathcal{X}_{ij}, j \in N(i)$
 server as **anchors**

Local Measurement Data $\mathcal{D}_i = \{(\mathbf{x}_{i,n}, \tilde{\Phi}_{i,n})\}_{n=1}^{N_i}$,

$\mathbf{x}_{i,n} \in \mathcal{X}_i$, $\tilde{\Phi}_{i,n} = \Phi(\mathbf{x}_{i,n}) + v_{i,n}$

UA-D-RME: Local Probabilistic Model



Encoder $E_{\phi_i}(\cdot)$:

$$(\mu_i, \ell_i) = E_{\phi_i}(X_i) \triangleq \rho_{\phi_i} \left(\frac{1}{N_i} \sum_{n=1}^{N_i} \varphi_{\phi_i}([E_f(\mathbf{x}_{i,n}), \tilde{\Phi}_{i,n}], e(\mathbf{x}_{i,n})) \right),$$

Define a diagonal Gaussian posterior distribution over the latent variable $\mathbf{z}_i$:

$$q_i(\mathbf{z}_i | X_i) = \mathcal{N}(\mathbf{z}_i; \mu_i, \text{diag}(\exp(\ell_i)))$$

Decoder $D_{\psi_i}(\cdot)$:

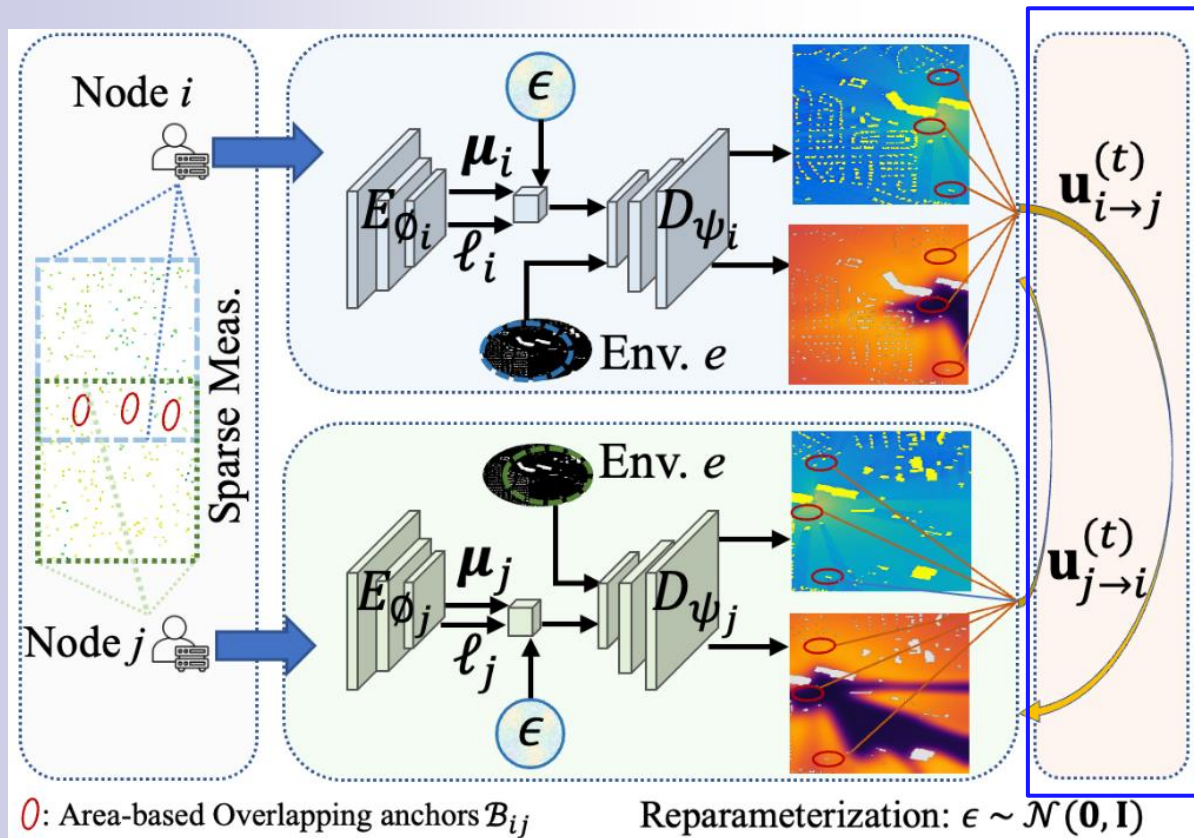
$$\hat{\Phi}_i(\mathbf{x}; \mathbf{z}_i) = D_{\psi_i}(\mathbf{x}, e(\mathbf{x}), \mathbf{z}_i), \quad \mathbf{x} \in \mathcal{X}_i$$

Yielding

$$\hat{\Phi}_i(\mathbf{x}) = \mathbb{E}_{\mathbf{z}_i \sim q_i} [D_{\psi_i}(\mathbf{x}, e(\mathbf{x}), \mathbf{z}_i)],$$

$$\hat{\mathbf{v}}_i(\mathbf{x}) = \text{Var}_{\mathbf{z}_i \sim q_i} [D_{\psi_i}(\mathbf{x}, e(\mathbf{x}), \mathbf{z}_i)] + \sigma_{\min}^2,$$

UA-D-RME: UA Statistical Consensus



$$\hat{\Phi}_{ij}^{(t)} = \left[\hat{\Phi}_{ij}^{(t)}(\mathbf{b}_{ij,1}), \dots, \hat{\Phi}_{ij}^{(t)}(\mathbf{b}_{ij,Q_{ij}}) \right]^\top,$$

$$\mathbf{v}_{ij}^{(t)} = \left[\mathbf{v}_{ij}^{(t)}(\mathbf{b}_{ij,1}), \dots, \mathbf{v}_{ij}^{(t)}(\mathbf{b}_{ij,Q_{ij}}) \right]^\top.$$

Statistical vectors

$$\mathbf{u}_{i \rightarrow j}^{(t)} = \left[\hat{\Phi}_{ij}^{(t)}; \mathbf{v}_{ij}^{(t)} \right] \in \mathbb{R}^{2Q_{ij}}$$

Loss of Uncertainty-Aware D-RME

- Precision-weighted mean consensus loss

$$\mathcal{L}_{\text{PWC},i}^{(t)} = \frac{1}{|\mathcal{B}_i|} \sum_{j \in \mathcal{N}(i)} \left\| \hat{\Phi}_{ij}^{(t)} - \bar{\Phi}_{ij}^{(t)} \right\|_{\bar{\Lambda}_{ij}^{(t)}}^2 \triangleq (\hat{\Phi}_{ij}^{(t)} - \bar{\Phi}_{ij}^{(t)})^T \bar{\Lambda}_{ij}^{(t)} (\hat{\Phi}_{ij}^{(t)} - \bar{\Phi}_{ij}^{(t)})$$

where

$$\bar{\Phi}_{ij}^{(t)} = (\Lambda_{ij}^{(t)} + \sum_{j \in \mathcal{N}(i)} \Lambda_{ji}^{(t)})^{-1} (\Lambda_{ij}^{(t)} \hat{\Phi}_{ij}^{(t)} + \sum_{j \in \mathcal{N}(i)} \Lambda_{ji}^{(t)} \hat{\Phi}_{ji}^{(t)}),$$

$$\bar{\Lambda}_{ij}^{(t)} = \Lambda_{ij}^{(t)} + \sum_{j \in \mathcal{N}(i)} \Lambda_{ji}^{(t)},$$

- Local supervised objective loss

$$\mathcal{L}_{\text{sup},i}^{(t)} = \sum_{\mathbf{x} \in \mathcal{X}_i} \left[\frac{\left\| \Phi^{(t)}(\mathbf{x}) - \hat{\Phi}_i^{(t)}(\mathbf{x}) \right\|^2}{2 \max(\hat{\mathbf{v}}_i^{(t)}(\mathbf{x}), \epsilon_{\mathbf{v}})} + \frac{1}{2} \log \max(\hat{\mathbf{v}}_i^{(t)}(\mathbf{x}), \epsilon_{\mathbf{v}}) \right].$$

- Regularized loss

$$\mathcal{L}_{\text{KL},i}^{(t)} = \text{KL}(q_i(\mathbf{z}_i^{(t)} \mid \mathcal{D}_i) \parallel \bar{\mathcal{N}}(\mathbf{0}, \mathbf{I}))$$

- Total local loss of node i

$$\mathcal{J}_i^{(t)}(\Theta_i) = \mathcal{L}_{\text{sup},i}^{(t)} + \beta \mathcal{L}_{\text{KL},i}^{(t)} + \lambda \gamma(t) \mathcal{L}_{\text{PWC},i}^{(t)},$$

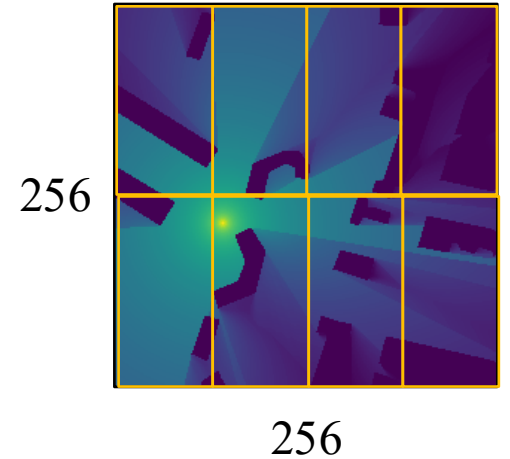
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Experiments of Uncertainty-Aware D-RME

□ Simulation Setup

- Target region: 1 Tx, 256*256 grids for 256m * 256m region
- Dataset:
 - ❖ 700 regions, each region can contribute 80 radio maps and the corresponding Tx.
 - ❖ The transmitter radiates signals at a power of 23dBm.
- DL system: a 256×256 spatial grid and partitioned into a 2×4 topology of overlapping sensing regions, monitored by $K = 8$ distributed mobile edge node



Experiments of Uncertainty-Aware D-RME

TABLE I

MAIN TEST RESULTS AVERAGED OVER THREE SEEDS. LOWER IS BETTER FOR RMSE, MAE, NMSE, NLL, AND MPIW. PICP IS EVALUATED AGAINST A 95% TARGET. DETERMINISTIC REFERENCES DO NOT PROVIDE VALID PREDICTIVE INTERVALS.

Method	RMSE↓	MAE↓	NMSE↓	NLL↓	PICP($\rightarrow$ 0.95)	MPIW↓
S-RME	4.57	3.43	0.0021	2.91	0.964	21.26
FedAvg	5.38	4.14	0.0030	3.13	0.966	25.89
EWMC	7.12	5.73	0.0051	4.30	0.801	23.53
PI-D-RME	5.86	3.94	0.0035	—	—	—
FedProx	6.17	4.75	0.0039	3.28	0.973	32.11
DP-SGD	5.81	4.43	0.0035	3.16	0.972	28.76
Dec-AE	4.38	3.32	0.0020	—	—	—
C-RME	5.21	4.05	0.0028	—	—	—
UA-D-RME	4.12	2.69	0.0017	64.92	0.936	15.16

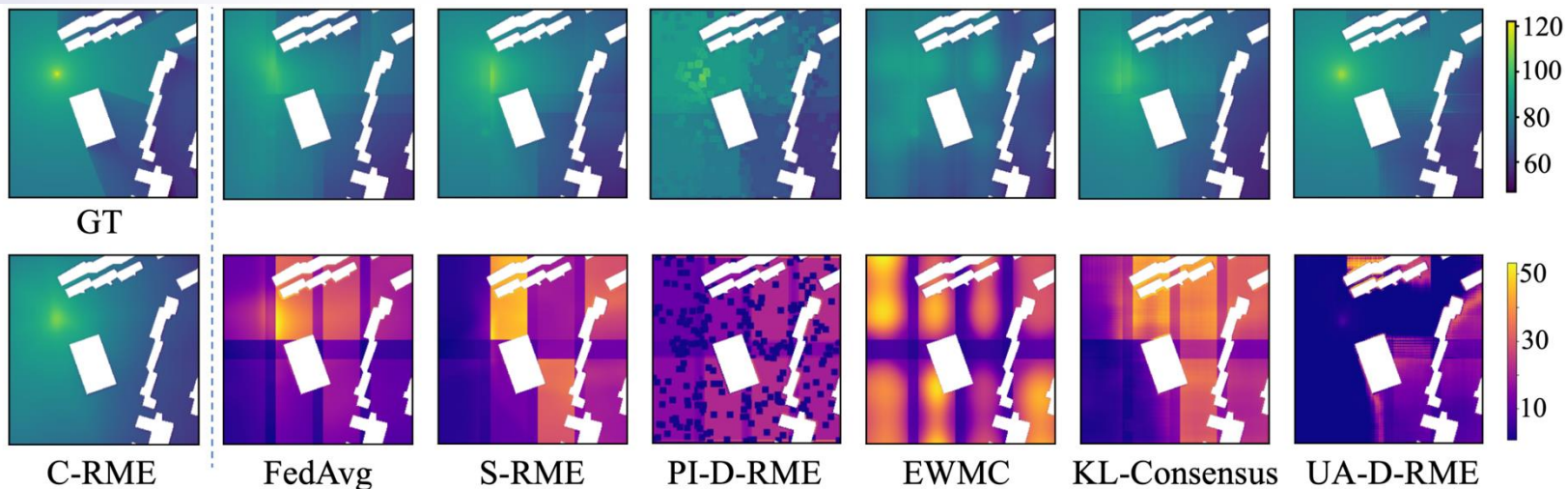


Fig. 4. Visualization of the inferred global radio map and the corresponding uncertainty map.

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Conclusion

✓ **Decentralized RME**

We formulate the communication-constrained RME problem in a decentralized manner. This formulation shifts RME paradigm from conventional centralized solutions toward decentralized, region-wise, autonomous coordination for efficient distributed RME.

✓ **Uncertainty-Aware Statistics Consensus**

Based on consensus optimization, each node iteratively reaches estimation consistency with neighbors at reduced communication cost than benchmark methods, by exchanging only uncertainty-aware radio statistics across the shared monitoring areas.

✓ **Empirical Validation**

We conduct experiments on urban radio map datasets to evaluate UA-D-RME, which achieves desired estimation accuracy comparable to a centralized RME benchmark at the reduced communication cost, while preserving data privacy.

Thank You!

